GURU GOBIND SINGH INDRAPRASTHA UNIVERSITY

B.TECH. COMPUTER SCIENCE & ENGINEERING / ALLIED BRANCHES

END SEMESTER EXAMINATION (2023-24) -- SEMESTER 3

COURSE: COMPUTER ORGANIZATION & ARCHITECTURE (ETCS-302)

Time Allowed: 3 Hours Maximum Marks: 70 Marks Paper Code: ETCS-302

Roll Number: [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] | Total Printed Pages: 2

NOTE: Attempt all Sections. Assume suitable missing data if any. Programmable calculators are NOT permitted.

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SECTION A: Short Answer Type Questions (Attempt ALL questions) [10 x 2 = 20 Marks]

Q1. (a) What is stored-program concept in Von-Neumann computer architecture?

Q1. (b) Define instruction cycle, fetch cycle, and execute cycle.

Q1. (c) What is micro-operation? Differentiate between hardwired and microprogrammed control units.

Q1. (d) Define spatial and temporal locality of reference in cache memory.

Q1. (e) What is bus arbitration? Differentiate between centralized and distributed arbitration.

Q1. (f) Calculate speedup factor in a 5-stage pipeline for 100 tasks.

Q1. (g) What is difference between memory-mapped I/O and isolated I/O?

Q1. (h) Define hit ratio and write formula for average memory access time (AMAT).

Q1. (i) What is pipeline hazard? Name three primary types of hazards.

Q1. (j) What is IEEE 754 single precision floating point representation format?

SECTION B: Analytical & Computational Problems (Attempt any THREE) [3 x 10 = 30 Marks]

Q2. (a) Multiply multiplicand (-9) by multiplier (-13) using Booth multiplication algorithm with 5-bit registers.

Q2. (b) Design a 4-bit carry lookahead adder and derive Boolean expressions for carry generation and propagation.

Q2. (c) A cache memory has 32 KB capacity, block size of 64 bytes. For 32-bit physical address, calculate tag, set, word bits for direct a

Q2. (d) Explain 5-stage instruction pipeline (IF, ID, EX, MEM, WB) and draw space-time diagram for 4 instructions.

Q2. (e) Explain DMA transfer modes (Burst, Cycle Stealing, Demand Transfer) with block diagram.

SECTION C: Long Technical & Design Questions (Attempt any ONE from each) [5 x 10 = 50 Marks]

Q3. (a) Explain hardwired control unit organization with state machine and decoder circuits. [10 Marks]

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(b) Explain microprogrammed control unit with control memory and microinstruction sequencing. [10 Marks]

Q4. (a) Explain non-restoring division algorithm with hardware architecture and trace 11 divided by 3. [10 Marks]

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(b) Explain floating point arithmetic operations (addition and multiplication) with flowchart. [10 Marks]

Q5. (a) Discuss data hazards, control hazards, and structural hazards in pipelining with forwarding solutions. [10 Marks]

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(b) Explain superscalar architecture, branch prediction, and speculative execution in modern processors. [10 Marks]

Q6. (a) Explain cache write policies: Write-through vs Write-back, Write-allocate vs No-write-allocate. [10 Marks]

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(b) Explain virtual memory architecture, page table address translation, and TLB operation. [10 Marks]

Q7. (a) Explain asynchronous data transfer using strobe control and handshaking mechanisms. [10 Marks]

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(b) Explain interrupt-driven I/O, daisy chain priority interrupt controller, and interrupt service routine. [10 Marks]

EXAMINATION EVALUATION SCHEME & RUBRIC GUIDELINES:

1. Section A: Direct definition, time complexity analysis, formula derivation, or syntax verification.
2. Section B: Step-by-step mathematical proof, trace tables, algorithmic pseudocode, and state diagrams.
3. Section C: Architectural design, full code implementation, edge case handling, and space/time tradeoffs.

***** END OF QUESTION PAPER -- DEVENGERS ACADEMIC VAULT *****